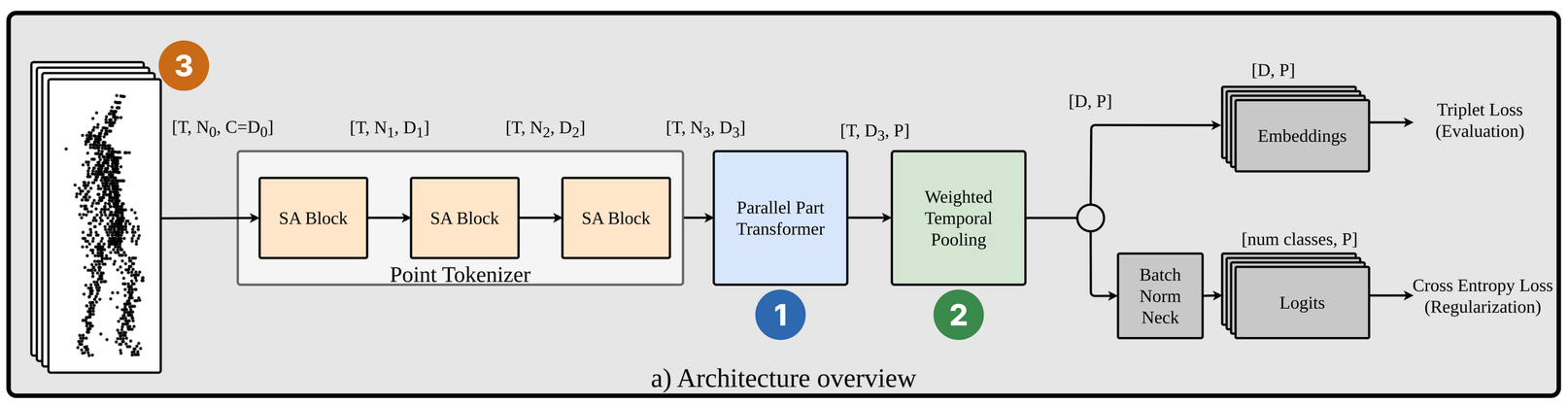
GaitPPT: Parallel Part-based Transformer for Gait Recognition from Lidar Point Clouds

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Method overview



a) Architecture overview

Three Set-Abstraction blocks turn each frame (1024 points) into 128 tokens (D = 256) that keep their height z. ① **PPT** reasons within height bins, ② **WTP** pools over the T frames, ③ **axis-decoupled scaling** augments the training clips. Loss: triplet + cross-entropy (OpenGait).

In a nutshell

Lidar gait encoders reason about body height explicitly only in their final pooling, and keep a single peak per channel over time. **GaitPPT** moves height into attention and learns *when* each channel matters.

94.23% Rank-1 on SUSTech1K · GaitPPT-B
+1.53 vs LidarGait++

85.74% Rank-1 on FreeGait · GaitPPT-L
+3.74 vs LidarGait++

2.87 M parameters · GaitPPT-B
34% fewer than LidarGait++

1 Parallel Part Transformer

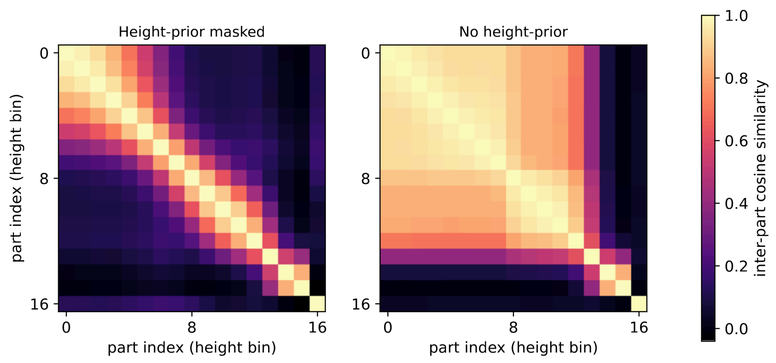
Gait is encoded along the vertical axis, so PPT builds that prior into attention itself.

- **Height mask:** a token attends only to tokens in its own height bin (absolute z, so size-aware).
- **One branch per scale 1/s:** a single layer each; same tokens, separate weights.
- **Parts:** per-bin mean + max pooling; scales {1/1, 1/16} give P = 17 part tokens.



Branches used: Tiny 1/1 · Base 1/1 + 1/16 · Large all five

Without the mask, parts collapse



Inter-part cosine similarity of the 17 part embeddings. With the mask, mean similarity drops **0.57 → 0.28** and the effective rank rises **6.5 → 8.7** (of 17).

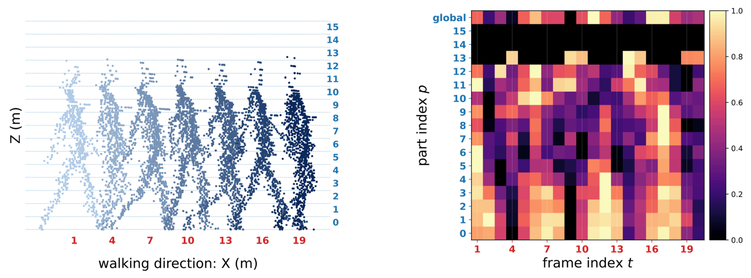
2 Weighted Temporal Pooling

Max-pooling keeps one frame per channel. WTP learns each channel's own weighting over time.

$$s = \tanh(x\,W_1)\,W_2 \in \mathbb{R}^{T\times D} \quad (1)$$

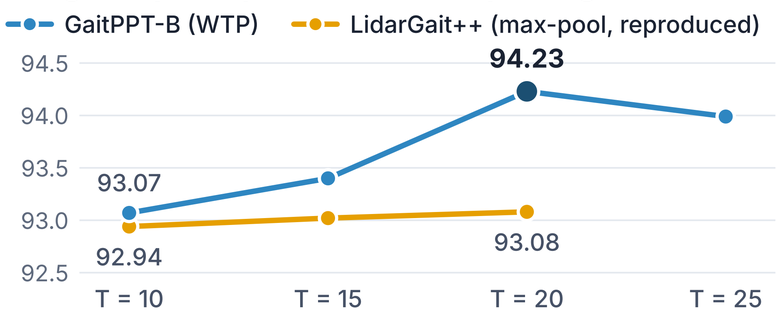
$$\hat{x}_d = \sum_{t=1}^{T} \mathrm{softmax}_t(s_{:,d})_t \; x_{t,d} \quad (2)$$

- Softmax over T, per channel and part; the output stays $D \times P$, a drop-in for max-pooling.
- Only $2D^2 \approx 131$ k extra parameters.



Channel-averaged attention per (part, frame) on a normal-walking clip; 1/16 height bins at left. Leg bins peak at gait landmarks such as the leg cross (t = 4); the global bin stays diffuse.

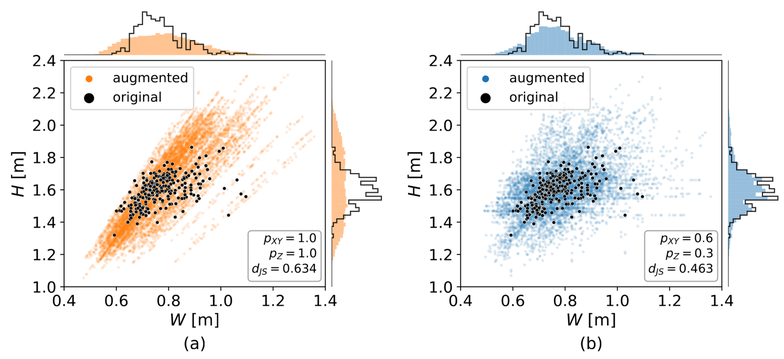
Longer clips help WTP, not max-pooling



3 Axis-decoupled scaling

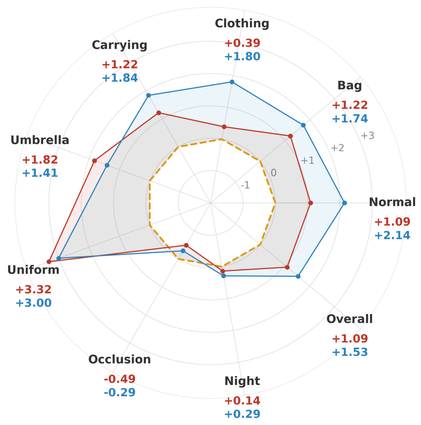
People vary far more in width than in height, so XY and Z get independent scale factors.

- XY and Z scaled by U[0.8, 1.25] with p = 0.6 and 0.3 (SUSTech1K); ≈ 28% of clips stay unscaled.
- Closer to real bodies than isotropic scaling: Jensen–Shannon distance **0.463** vs. **0.634**.



Body width W vs. height H: 250 training subjects (black), 40 augmentations each; (a) isotropic (LidarGait++), (b) ours.

Rank-1 gain over LidarGait++ per condition



Clothing (hardest)
+1.80 R1
uniform scaling: +0.39

Normal walking
+2.14 R1
uniform scaling: +1.09

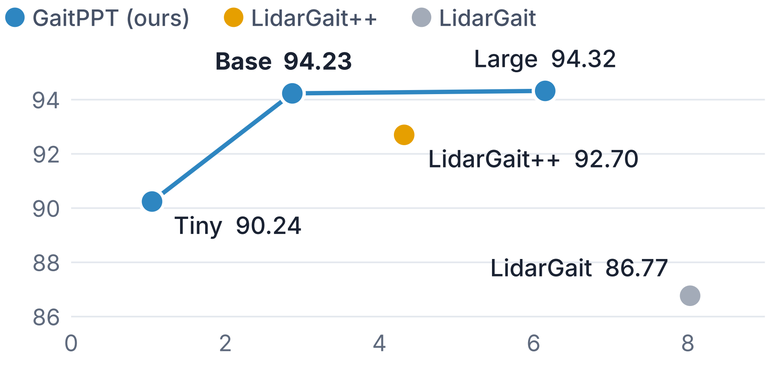
decoupled (ours)
uniform scaling
LidarGait++ baseline

Results

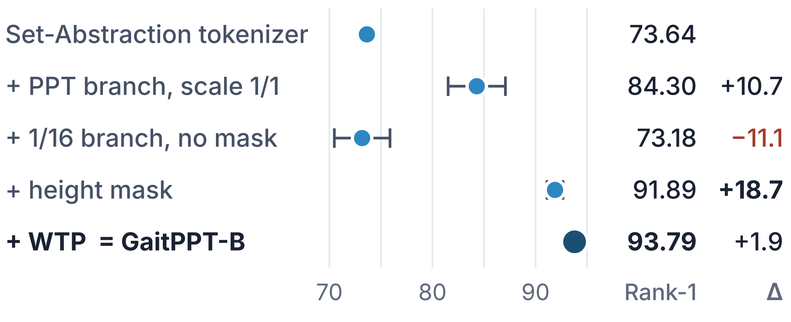
SUSTech1K	Input	Params	NM	BG	CL	CR	UB	UF	OC	NT	Overall
LidarGait	depth	8.03 M	91.80	88.64	74.56	89.03	67.50	80.86	94.53	90.41	86.77
HMRNet	depth + points	–	92.70	92.30	79.60	90.30	83.10	86.20	95.20	90.40	90.20
GaitCloud	voxels	–	89.82	90.30	79.17	89.65	85.79	89.02	94.39	90.62	89.20
LidarGait++	points	4.32 M	94.20	93.90	79.70	92.40	91.50	91.90	96.60	92.20	92.70
GaitPPT-Tiny	points	1.05 M	94.11	91.44	76.01	89.59	87.79	90.89	94.82	89.19	90.24
GaitPPT-Base	points	2.87 M	96.34	**95.64**	**81.50**	94.24	92.91	94.90	96.31	**92.49**	94.23
GaitPPT-Large	points	6.15 M	**96.84**	95.59	81.08	**94.35**	**93.01**	**94.93**	**96.69**	92.35	**94.32**

Cross-view Rank-1 (%), single run, bold = best · NM normal · BG bag · CL clothing · CR carrying · UB umbrella · UF uniform · OC occlusion · NT night

SUSTech1K · Rank-1 (%) vs. parameters (M)



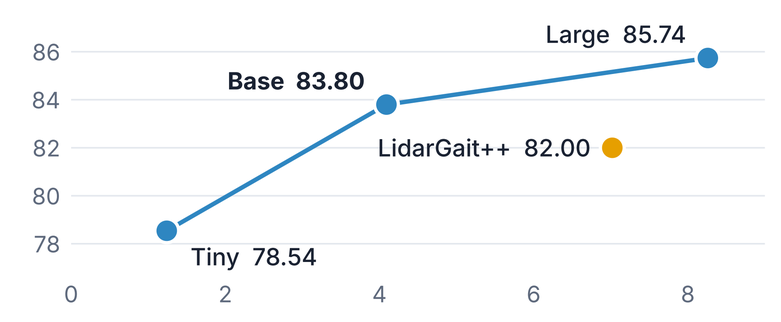
Every component counts 5 seeds, mean ± std



FreeGait outdoor · zero-shot = trained on SUSTech1K

FreeGait	Params	Rank-1	mAP	Zero-shot
LidarGait	–	74.20	80.70	**23.00**
HMRNet	–	80.80	86.50	–
LidarGait++	7.02 M	82.00	87.20	15.10
GaitPPT-Tiny	1.24 M	78.54	84.30	17.22
GaitPPT-Base	4.09 M	83.80	88.45	19.86
GaitPPT-Large	8.26 M	**85.74**	**89.80**	18.73

FreeGait · Rank-1 (%) vs. parameters (M)



References. LidarGait: Shen et al., CVPR 2023 · LidarGait++: Shen et al., CVPR 2025 · FreeGait / HMRNet: Han et al., ACM MM 2024 · GaitCloud: Zhang et al., WACV 2025 · PointNet++: Qi et al., NeurIPS 2017 · OpenGait: Fan et al., CVPR 2023

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Paper, code & video
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